EVENTSYNC AI

Agentic Event Volunteer Coordination Platform

A Mini Project Report Submitted in Partial Fulfillment of the Requirements for the Award of the Degree of

BACHELOR OF TECHNOLOGY

in

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

We hereby declare that the mini project report entitled “EventSync AI – Agentic Event Volunteer Coordination Platform” is a record of bonafide work carried out by us in partial fulfillment of the requirements for the award of the Bachelor of Technology degree in Computer Science and Engineering at K L (Deemed to be) University during the academic year 2026–2027. The work presented in this report is our own and has not been submitted to any other department, university, or institute for the award of any degree or diploma.

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CERTIFICATE

This is to certify that the mini project report entitled “EventSync AI – Agentic Event Volunteer Coordination Platform” is a bonafide work carried out and submitted by P. Sri Maha Lakshmi (2420030313), B. Nageshwari (2420030426), and R. Ananya (2420030086), in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering at K L (Deemed to be) University during the academic year 2026–2027.

Signature of the Course Co-ordinator **Signature of HOD**

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1. ABSTRACT

Volunteer coordination is a critical operational challenge at large-scale events. Organizers must assign volunteers to tasks according to skills, availability, experience, and current workload while responding to last-minute cancellations. Manual allocation using spreadsheets, calls, and personal judgement can be time-consuming and may lead to mismatched assignments, uneven workloads, and delays.

EventSync AI is a proposed agentic event volunteer coordination platform that applies Service-Oriented Architecture (SOA) and microservices to streamline this process. The platform is organized around Event, Volunteer, Task/Assignment, and Notification services, supported by an AI Event Coordination Agent. The agent evaluates event requirements and volunteer profiles to recommend suitable assignments, identify replacements when a volunteer becomes unavailable, and balance work across the available volunteer pool.

The proposed system uses Java and Spring Boot for backend services, Spring Cloud and Eureka for service discovery, Spring Cloud Gateway for routing, REST APIs for communication, and React for the frontend. PostgreSQL or MySQL is proposed for data persistence. JWT authentication and role-based access control are included in the planned security design. Implementation and evaluation are intended to assess the system's usefulness in reducing manual coordination and improving assignment responsiveness.

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2. INTRODUCTION

Service-Oriented Architecture (SOA) is a software design approach in which an application is organized as a collection of independent, loosely coupled services that communicate through well-defined interfaces. Microservices apply this principle by separating application capabilities into independently developed and maintained components.

EventSync AI applies SOA and microservices to event volunteer coordination. The proposed platform separates event management, volunteer profiles, task assignments, and notifications into dedicated services. An AI Event Coordination Agent provides the coordination logic that evaluates skills, availability, experience, and workload when recommending assignments.

When a volunteer becomes unavailable, the system is intended to identify a suitable replacement, update the assignment, and notify affected parties. Workload balancing is also included to help avoid repeatedly assigning too many tasks to the same volunteers. The design aims to improve modularity, maintainability, and scalability while reducing the manual effort required from event organizers.

Project Overview

Field	Details
Project Title	EventSync AI – Agentic Event Volunteer Coordination Platform
Course	SOA Programming and Microservices (24SDCS03)
Academic Year	2026–2027
Department	Computer Science and Engineering, KLH CSE Bowrampet Campus
Project Guide	Dr. Neelima Gogineni
Architecture	Service-Oriented Architecture (SOA) using microservices

3. LITERATURE SURVEY

3.1 Sudin et al. — Preference Matching Algorithm

Methodology: The referenced work describes a web-based volunteer scheduling system using preference-based matching aligned with volunteer preferences, skills, and experience.

Inference: The approach can streamline organizer scheduling and support fairer job distribution, with evaluation based on match rate and average preference satisfaction.

Limitation: The cited project focuses on sports-event volunteers and does not address real-time reassignment when a volunteer becomes unavailable.

3.2 Samanta and Bhattyacharya — LLM-Assisted Task Assignment

Methodology: The work combines large language model (LLM)-assisted semantic skill extraction with a utility-driven assignment engine that considers skills and willingness, alongside a blockchain-based execution framework.

Inference: The project documentation reports improvements in assignment utility and task coverage compared with skill-only greedy matching.

Limitation: LLM and blockchain components can add overhead, and the work does not specifically focus on dynamic reassignment following volunteer dropout.

3.3 Haider et al. — Fault-Tolerant Dynamic Task Reassignment

Methodology: The referenced work combines real-time failure detection with recovery-driven task reassignment and MQTT-based cloud coordination between agents.

Inference: The study reports reduced task completion time and downtime compared with baseline scheduling approaches as system load increases.

Limitation: The work is applied to multi-robot systems rather than human volunteers; further work on machine-learning-based task prediction is identified.

3.4 Summary

The reviewed work highlights preference-based matching, semantic task assignment, and dynamic recovery as relevant directions. EventSync AI proposes to bring multi-criteria assignment, volunteer unavailability handling, workload balancing, and service-based architecture together for event volunteer coordination.

4. PROBLEM STATEMENT AND OBJECTIVES

4.1 Problem Statement

To design and develop an SOA-based event volunteer coordination platform that assists organizers in allocating volunteers to tasks, responding to volunteer unavailability, and balancing workloads through an AI Event Coordination Agent.

4.2 Problem Description

- Volunteer coordination is often performed manually using spreadsheets, phone calls, and personal judgement.
- Organizers may find it difficult to match skills, experience, availability, and workload with task requirements.
- Last-minute volunteer cancellations can leave tasks uncovered while a replacement is identified manually.
- Without a unified coordination mechanism, workload may be unevenly distributed and updates may not reach affected parties promptly.

4.3 Objectives

1. Design a microservices-based volunteer management system with Event, Volunteer, Task/Assignment, and Notification services.
2. Develop an AI Event Coordination Agent that recommends assignments using skills, availability, experience, and workload.
3. Support dynamic reassignment when a volunteer cancels or becomes unavailable.
4. Update assignment records and trigger notifications following reassignment.
5. Apply REST APIs, Eureka service discovery, Spring Cloud Gateway, JWT authentication, and role-based access control in the planned architecture.
6. Support Sustainable Development Goal 11 (Sustainable Cities and Communities) by improving the coordination of community and civic events.

4.4 Problem Gap

- Existing coordination tools may lack intelligent criteria-based volunteer allocation.
- Volunteer dropout and reassignment are often handled manually, causing delays.
- Workload balancing may be limited, creating overburdened or underused volunteers.
- Monolithic designs can make systems harder to extend and integrate with an AI coordination layer.

5. HARDWARE AND SOFTWARE REQUIREMENTS

5.1 Hardware Requirements

- Processor: Intel Core i3 or above.
- RAM: Minimum 4 GB; 8 GB recommended.
- Storage: At least 10 GB of free disk space.

5.2 Software Requirements

Component	Proposed Technology
Operating System	Windows 10/11 or Linux
Backend	Java, Spring Boot
Service Discovery	Spring Cloud, Eureka
API Gateway	Spring Cloud Gateway
Communication	REST APIs
Frontend	React
Database	PostgreSQL or MySQL
Security	JWT authentication, RBAC, CORS policies
API Testing	Postman Agent Mode
Security Scanning	Semgrep
CI/CD	GitHub Actions
Project Management	Jira
Design and Development Support	ChatGPT/Gemini, GitHub Copilot, Lucidchart AI

5.3 Feasibility Analysis

Technical feasibility: The proposed design uses established technologies such as Java, Spring Boot, Spring Cloud, React, and PostgreSQL/MySQL, with supporting documentation and developer communities.

Tooling feasibility: API testing, security scanning, CI/CD, and sprint tracking are planned with Postman Agent Mode, Semgrep, GitHub Actions, and Jira.

Team feasibility: The project is scoped around REST API design and microservices concepts, with AI-assisted development tools available to support design and coding.

Economic feasibility: The design relies mainly on open-source frameworks and free-tier development tools, making it suitable for an academic project.

6. IMPLEMENTATION

The following implementation describes the proposed design and planned workflow. The project documentation identifies these components and technologies; this report does not claim that all components have already been implemented or tested.

6.1 System Architecture

EventSync AI follows a Service-Oriented Architecture based on microservices. Four core services manage event data, volunteer profiles, assignments, and notifications. A dedicated AI Agent layer coordinates assignment decisions and workload balancing. Spring Cloud Gateway is the planned single entry point for client requests, while Eureka supports service discovery.

Component / Service	Responsibility
Event Service	Manages event creation, scheduling, and event-level metadata.
Volunteer Service	Manages volunteer registration, profiles, skills, availability, and workload.
Task/Assignment Service	Manages task definitions, volunteer-to-task assignments, and reassignment workflows.
Notification Service	Delivers notifications for assignments, reassignments, and event updates.
AI Agent Layer	Coordinates allocation logic, dynamic reassignment, and workload balancing.
API Gateway	Routes client requests to the appropriate service and supports centralized cross-cutting concerns.
Eureka Server	Enables service registration and discovery.

6.2 Allocation Logic

The AI Event Coordination Agent considers event requirements and volunteer profiles. The intended matching process evaluates multiple criteria together rather than relying on one attribute.

7. Read the event and task requirements.
8. Review volunteer skills, experience, availability, and current workload.
9. Compare available volunteers against task requirements.
10. Recommend suitable volunteer-to-task assignments.

6.3 Dynamic Reassignment

11. Detect or receive a volunteer unavailability or cancellation update.
12. Identify a suitable replacement from the remaining available volunteers based on skill match, availability, and workload.
13. Update the assignment record through the Task/Assignment Service.
14. Trigger a notification to the affected organizer and volunteer(s).

6.4 Workload Balancing

The agent is intended to consider current workload while recommending assignments so that tasks are distributed more evenly and individual volunteers are not repeatedly overburdened.

6.5 API Gateway, Security, and Persistence

- Spring Cloud Gateway serves as the planned single entry point and routes requests to relevant microservices.
- JWT-based authentication is planned to secure API interactions.
- Role-Based Access Control (RBAC) is intended to restrict actions according to roles such as organizer, volunteer, and administrator.
- CORS policies govern cross-origin requests from the React frontend.
- PostgreSQL or MySQL is proposed for storing event, volunteer, task, and assignment data. Each service owns its domain data in the database-per-service design.

6.6 Frontend

The proposed React interface allows organizers to manage events and assignments, while volunteers can view assigned tasks and update availability. The frontend communicates with backend services through REST APIs routed via the gateway.

7. RESULTS AND DISCUSSION

At the proposal/documentation stage, the expected result is a modular event volunteer coordination platform that brings event management, volunteer profiles, task assignment, notifications, and AI-assisted coordination into one system. The intended workflow supports initial allocation, reassignment after a cancellation, and workload-aware recommendations.

7.1 Expected Results

- Reduced manual effort in coordinating volunteers and event tasks.
- Faster identification of replacement volunteers when cancellations occur.
- More consistent assignment decisions based on skills, availability, experience, and workload.
- Improved visibility of assignments and reassignment updates through notifications.
- A modular architecture that can be extended service by service.

7.2 Evaluation Plan

After implementation, the system can be evaluated using representative event and volunteer scenarios. Useful measures include task coverage, time required to identify a replacement, assignment suitability, workload distribution, and notification delivery. Actual values should be added only after testing.

7.3 Discussion

The architecture separates business responsibilities into individual services and places coordination logic in a dedicated AI Agent layer. This separation is intended to support maintainability and future scaling. The effectiveness of allocation and reassignment will depend on the quality of volunteer profile data, the matching logic, and testing under realistic cancellation and workload scenarios.

8. CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

EventSync AI proposes an intelligent approach to event volunteer coordination by combining SOA-based microservices with an AI Event Coordination Agent. The system is designed to address manual allocation, delayed replacement of unavailable volunteers, and uneven workload through multi-criteria assignment, dynamic reassignment, and notification workflows. The proposed architecture includes Event, Volunteer, Task/Assignment, and Notification services, supported by Eureka, Spring Cloud Gateway, REST APIs, and a React frontend. Implementation and evaluation remain necessary to verify the expected benefits.

8.2 Future Scope

- Complete and integrate the core microservices and their REST endpoints.
- Implement and evaluate the AI-assisted allocation and workload-balancing logic.
- Add monitoring and stronger recovery handling for service failures.
- Enhance the frontend with live assignment status and organizer dashboards.
- Deploy the platform through a CI/CD pipeline and evaluate it using larger event scenarios.
- Explore additional AI methods for improving assignment quality and predicting staffing needs.

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